

Irrationality of $\sqrt{7}$

Problem. Prove that $\sqrt{7}$ is irrational.

Solution

Proof. Write $m \mid n$ to mean that $n = mk$ for some $k \in \mathbb{Z}$. We first prove the divisibility fact needed in the argument.

Lemma. If $a \in \mathbb{Z}$ and $7 \mid a^2$, then $7 \mid a$.

Proof of the lemma. By the division algorithm, there exist $k \in \mathbb{Z}$ and $r \in \{0, 1, 2, 3, 4, 5, 6\}$ such that

$$a = 7k + r.$$

Thus $a - r = 7k$, so $7 \mid (a - r)$. Hence

$$a^2 - r^2 = (a - r)(a + r)$$

is divisible by 7, which means $a^2 \equiv r^2 \pmod{7}$. The possible residues of r^2 modulo 7 are

r	0	1	2	3	4	5	6
$r^2 \pmod{7}$	0	1	4	2	2	4	1

so $r^2 \equiv 0 \pmod{7}$ occurs only when $r = 0$. Therefore, if $7 \mid a^2$, then $a^2 \equiv 0 \pmod{7}$, so $r^2 \equiv 0 \pmod{7}$, forcing $r = 0$. Hence $a = 7k$, so $7 \mid a$. This proves the lemma.

Now suppose, for contradiction, that there exist integers P and Q with $Q \neq 0$ such that

$$\left(\frac{P}{Q}\right)^2 = 7.$$

Let $d = \gcd(P, Q) > 0$, and define

$$p = \frac{P}{d}, \quad q = \frac{Q}{d}.$$

Then $p, q \in \mathbb{Z}$, $q \neq 0$, and $\frac{p}{q} = \frac{P}{Q}$. Also $\gcd(p, q) = 1$: indeed, if a positive integer c divided both p and q , then cd would divide both P and Q , so $cd \leq d$ by the definition of $d = \gcd(P, Q)$, forcing $c = 1$.

Thus

$$\left(\frac{p}{q}\right)^2 = 7$$

with $\gcd(p, q) = 1$. Since $q \neq 0$, multiplying both sides by q^2 gives

$$p^2 = 7q^2.$$

Therefore $7 \mid p^2$. By the lemma, $7 \mid p$, so there exists $s \in \mathbb{Z}$ such that

$$p = 7s.$$

Substituting this into $p^2 = 7q^2$, we get

$$49s^2 = 7q^2.$$

Dividing by 7 gives

$$q^2 = 7s^2.$$

Hence $7 \mid q^2$. Applying the lemma again, $7 \mid q$.

Thus 7 divides both p and q , contradicting $\gcd(p, q) = 1$. Therefore no integers P and Q with $Q \neq 0$ satisfy

$$\left(\frac{P}{Q}\right)^2 = 7.$$

Hence $\sqrt{7}$ is irrational. □